VASP Analyzer

User Guide

A desktop application for reading, visualising and editing VASP calculations. Open a calculation folder and every page configures itself from what is actually there — a page whose data is missing says which file it needs rather than failing.

Opening a calculation

Three ways in:

- **Drag and drop** a folder onto the window.
- **File → Open Calculation…** (Ctrl+O) for a folder.
- **File → Open File…** (Ctrl+Shift+O) for a single file.

A **flat folder** holding VASP files directly works as-is. A **project folder** organised into band, md, neb and phonon subdirectories also works — each page family resolves to its own subfolder independently, so a project with only some of them is fine. Dropping an md or neb folder onto an already-open project attaches just that part and leaves the rest untouched.

Single files work too. A lone **POSCAR** opens the Structure and Editor pages; a lone **CHGCAR** opens those plus Charge Density, because a volumetric file carries a full structure in its header. Everything else shows its ordinary "not found" state naming what is missing.

The window

- **Left sidebar** — display controls for the page you are on.
- **Centre** — the page itself.
- **Top right** — Reset, Copy, Export, and a ⋯ menu (Open, Reload, Reveal in Finder, Show Log).
- **Bottom bar** — key facts about the calculation; the fields change per page.

Pop out any plot or viewer with the ↗ button in its corner. It opens in its own resizable window with Copy and Export buttons, and stays live — the same interactive widget, not a picture. Close it to put it back.

Reload Calculation restarts the application. You will be asked to confirm, because unsaved Editor changes are lost.

Keyboard shortcuts

Global

Key	Action
`Ctrl+1 … Ctrl+8`	Jump to a page (Structure → Inputs)
`Ctrl+O`	Open a calculation folder
`Ctrl+Shift+O`	Open a single file
`F5`	Reload calculation (restarts the app; asks first)

3D viewers

Available on Structure, Editor, Charge, MD and NEB.

Key	Action
`S`	Select tool
`R`	Rotate the camera
`T`	Pan / translate the camera
`M`	Measure (Structure and Editor only)
`+ / =`	Zoom in
`–`	Zoom out
`F`	Fit the structure in view
`Middle-drag / Shift+drag`	Pan from any tool
`Scroll`	Zoom
`Double-click`	Reset the view
`Ctrl / Shift + click`	Add or remove atoms from the selection (MD, NEB)

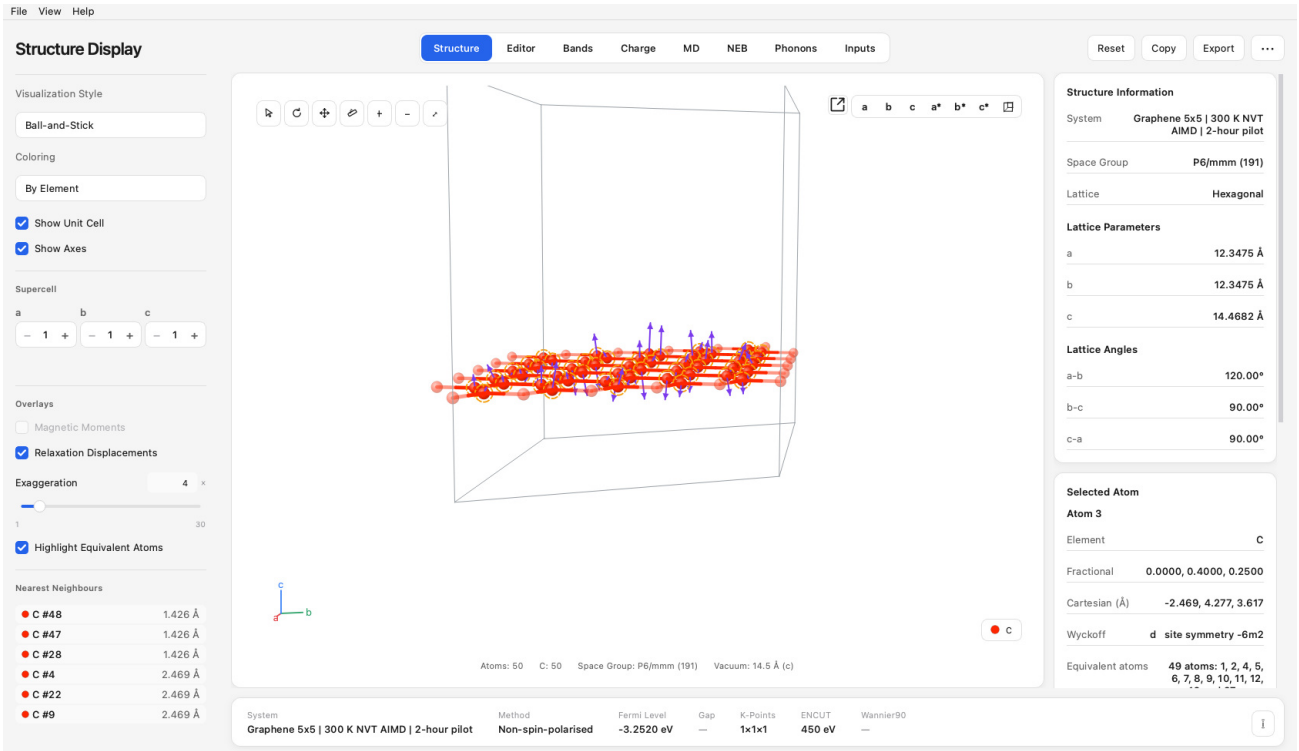
Editor

Key	Action
`G`	Grab and move the selection
`D`	Duplicate the selection
`H`	Hide the selected atoms
`Shift+H`	Isolate — hide everything else
`Alt+H`	Show all hidden atoms
`Delete / Backspace`	Delete the selection
`Cmd+Z`	Undo
`Cmd+Shift+Z`	Redo
`Esc`	Back to the Select tool

On Windows and Linux, Cmd is Ctrl and Redo is Ctrl+Y.

Structure

Read a crystal structure: geometry, symmetry and neighbours.



The Structure page, showing relaxation displacement arrows and the atoms symmetry-equivalent to the selected one.

Opens on the POSCAR (or CONTCAR when no POSCAR exists). The viewer shows the unit cell with bonds inferred from covalent radii; the right panel reports the lattice parameters, angles and the space group determined by **spglib**.

What you can do

- **Supercell** — replicate the cell along a, b and c with the +/- steppers. The atom count is shown before you commit, so a 6×6×6 that would produce tens of thousands of atoms is visible as such first.
- **Measure** — click two atoms for a distance, three for an angle. Right-click clears an in-progress measurement. Offered here and in the Editor only: a distance describes one fixed geometry, which is ambiguous on a trajectory.
- **Nearest neighbours** — select an atom to list its closest contacts with distances.
- **Element colours** — click a legend swatch to override any element's colour; the change applies everywhere in the app.
- **Slab awareness** — when a vacuum gap is detected the camera frames the atoms rather than the mostly-empty cell, and the gap is reported.

Overlays

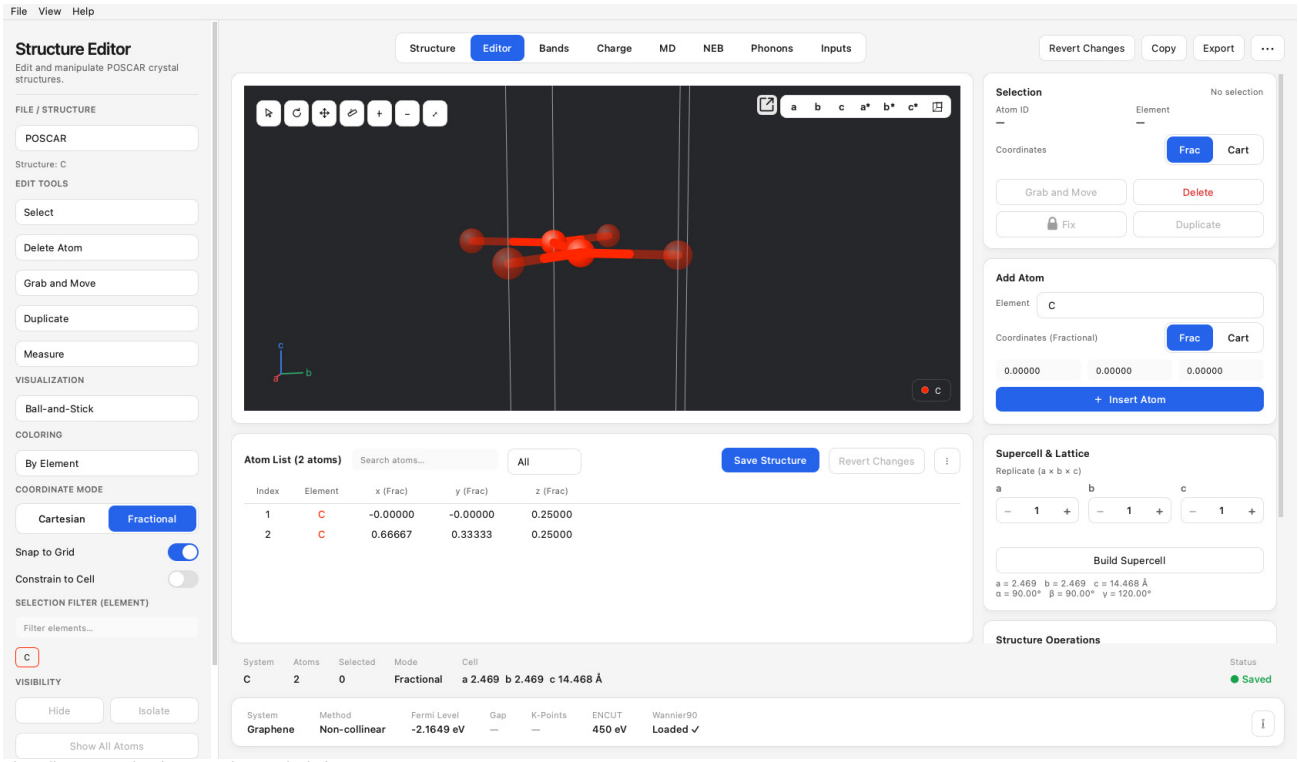
- **Relaxation Displacements** — with both a POSCAR and a CONTCAR present, draws an arrow on every atom from where it started to where the relaxation left it, using the minimum-image convention so an atom that crossed a cell face is not shown as a huge move. The *Exaggeration* slider scales the arrows for visibility only; the *Relaxation* card reports the true mean and RMS displacement and lists the most-displaced atoms. Shown only when atoms actually moved -- a static SCF or bands run copies POSCAR to CONTCAR unchanged, and there the toggle stays disabled with that explanation.
- **Magnetic Moments** — for a spin-polarised run with LORBIT set, draws each atom's moment from OUTCAR's final magnetization table. Collinear moments point along z, coloured by sign so antiferromagnetic order reads as alternating colour. **Non-collinear runs** (LNONCOLLINEAR / LSORBIT) carry a full spin vector per atom: the selector beneath the checkbox switches between the whole vector and S_x, S_y or S_z alone, so a canted or spiral texture can be read one axis at a time, and the selected atom's card lists all three components. The *Magnetization* card separates the **net moment** (the vector sum) from **Σ|m|** (the sum of magnitudes) — an antiferromagnet has a large Σ|m| and a net of nearly zero, and quoting only one of the two would describe it wrongly. A run whose moments all converged to zero (MAGMOM = 0 everywhere) keeps the card but disables the arrows, since there is nothing to draw.
- **Highlight Equivalent Atoms** — selecting an atom rings every atom the space group maps it onto, and the atom card reports its Wyckoff letter and site symmetry. Answers "which atoms are actually distinct here?" in one click.

Overlays whose data is absent (no CONTCAR, a non-magnetic run) are shown disabled with the reason in their tooltip rather than drawing nothing. All three follow a supercell correctly.

Clicking a file in the *Input Files* list jumps to it on the Inputs page.

Editor

Build and modify structures, then export them.



The Editor page, showing a graphene calculation.

A full structure editor with undo/redo. Everything is a working copy — nothing touches your files until you use **Save Structure** or one of the export actions. The bottom bar shows the atom count, selection, coordinate mode and a green *Saved* / amber *Modified* state at all times.

Editing

- **Select** atoms in the viewport or the table; the two selections stay in step. Shift-click extends. The element filter chips restrict what a click can select, which is what makes a dense multi-element cell workable.
- **Grab and Move** (G) drags the selection along a coloured axis handle, or toward a neighbouring atom. Handles are sized in screen pixels, so they stay grabbable at any zoom.
- **Delete**, **Duplicate** (D) and **Fix** (selective dynamics) act on the whole selection.
- **Add Atom** inserts at an exact fractional or Cartesian coordinate.
- **Wrap**, **Center**, **Sort by element** and **Check overlaps** are one-click structure operations.

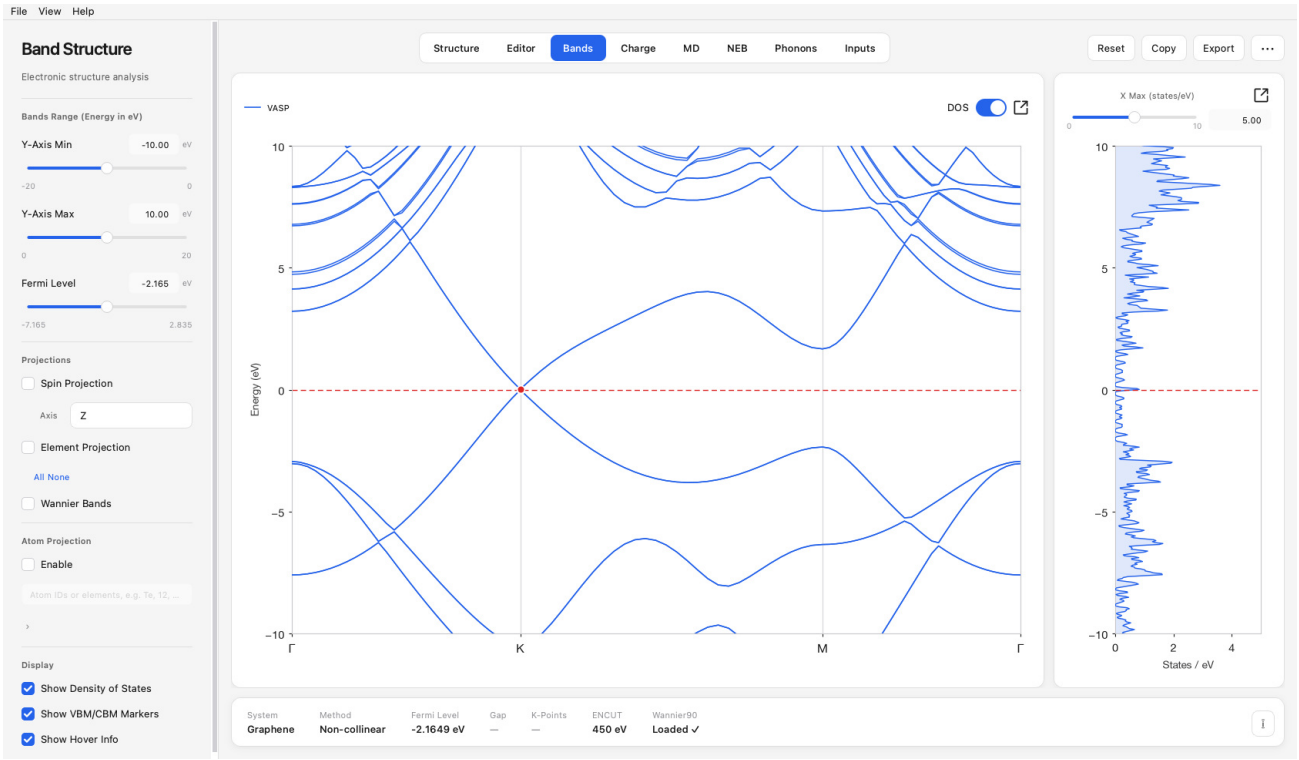
Visibility

Hide (H) removes the selected atoms from view, **Isolate** (Shift+H) hides everything else, and **Show All Atoms** (Alt+H) restores them. Hiding is a view state only — hidden atoms stay in the structure, in the table, and in anything you save. The sidebar always reports how many are hidden.

Export formats: POSCAR, CIF and XYZ.

Bands

Band structure and density of states, side by side.



The Bands page, showing a graphene calculation.

Reads EIGENVAL for the bands and DOSCAR for the density of states. Either works without the other — a DOSCAR with no EIGENVAL still draws the DOS. High-symmetry labels come from KPOINTS when it is in line mode; they are never invented.

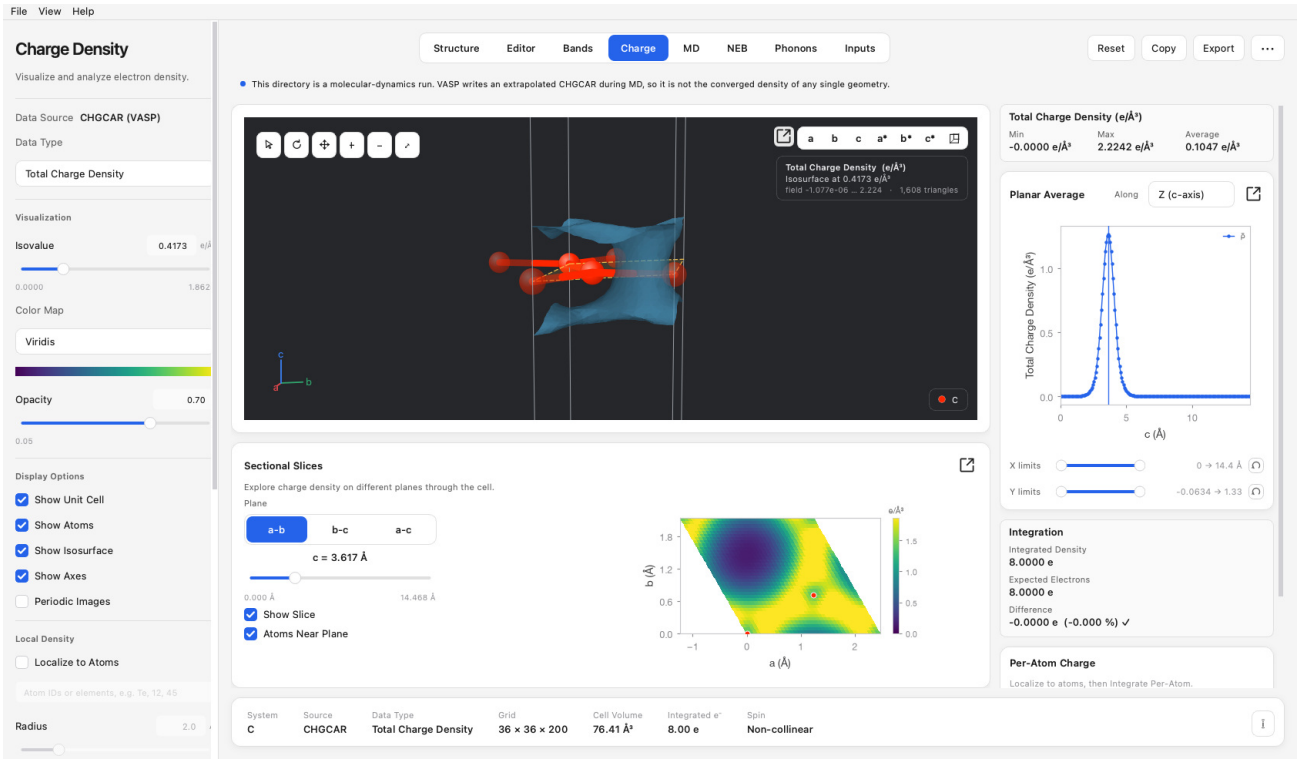
Reading the plot

- **Band gap** is computed and reported in the bottom bar, with VBM and CBM marked on the plot when enabled.
- **Hover** anywhere on the bands to read the k-point, band index, energy and orbital character under the cursor. Toggle it off with *Show Hover Info*.
- **Projections** — with PROCAR present, colour the bands by element or by orbital (s, p, d, and individual $p_x/p_y/p_z$ …). Line width can encode projection weight ("fatbands").
- **Spin** — spin-polarised runs draw both channels; the control is disabled with an explanation when the calculation has only one.
- **Wannier bands** overlay when an interpolated wannier90_band.dat is present.

Paper export writes a single-column, publication-sized PDF or SVG of the band+DOS pair with smaller fonts and tightened margins — not a screenshot.

Charge

Real-space volumetric fields: isosurfaces, slices and integrals.



The Charge page, showing a graphene calculation.

Reads CHGCAR, CHG, PARCHG, ELFCAR and LOCPOT. The *Data Source* picker lists every volumetric file found. Everything on the page derives from one selected field, so no panel can show a previous field's numbers.

Views

- **Isosurface** — a 3D surface at your chosen isovalue, with adjustable opacity, smoothing and subsampling. Periodic images can be drawn around the cell.
- **Sectional slice** — a 2D cut through the a-b, b-c or a-c plane, positioned anywhere along the normal. Non-orthogonal cells are drawn as the real parallelogram, not squared off.
- **Planar average** — the field averaged over each plane, plotted along the axis. The standard way to read a work function out of a LOCPOT.

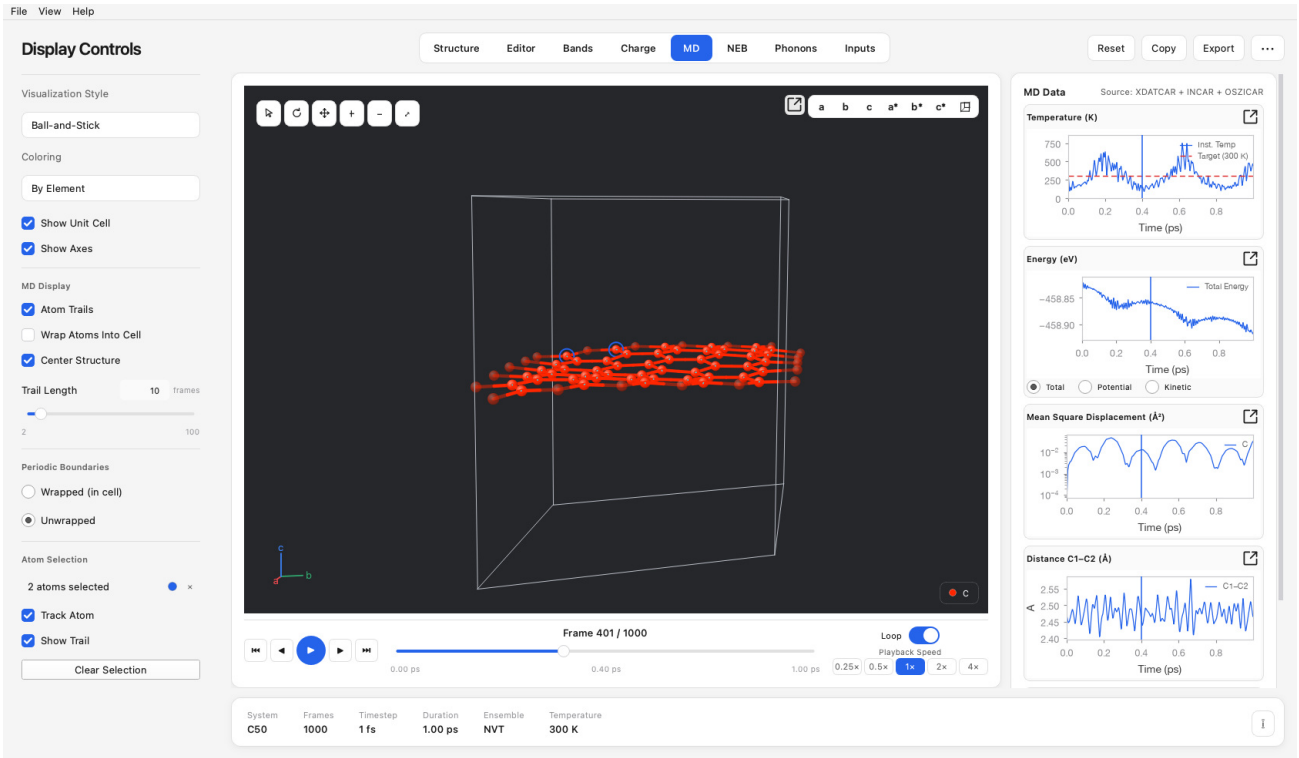
Analysis

- **Integration** — the whole-cell integral, which for a CHGCAR should return the valence electron count (NELECT). Shown alongside the expected value as a check.
- **Localize to Atoms** — zero the density beyond a radius of the atoms you name (by element or index), so the isovalue, statistics and plots describe only that region.
- **Integrate Per-Atom** — the integral inside that radius for each selected atom individually. A cheap approximation for comparing atoms of the same element in different environments.
- **Charge Difference** — subtract a second CHGCAR to visualise charge transfer. The colour map switches to a diverging one automatically, since the result is signed.

Per-atom integration is not Bader analysis. It sums a sphere, so overlapping spheres double-count and nothing is assigned by zero-flux surface. CHGCAR carries no true per-atom decomposition.

MD

Play back a molecular-dynamics trajectory against its telemetry.



The MD page with two atoms selected: their distance over the run appears as the fourth chart.

Reads XDATCAR for the trajectory and OSZICAR for temperature and energies. The viewer and the charts are two views of one state — the current frame — and scrubbing either moves both.

Playback

Transport controls step, play and scrub; playback speed is separate from the physical timestep, so a 0.5 fs run does not play at half the speed of a 1 fs one. Click any chart to seek to that time.

Viewer

- **Atom trails** draw where atoms have recently been, fading with age so the direction of travel reads without an arrowhead. Trails use one distinct colour rather than the element colour, so they never blend into the structure.
- **Selection** — click an atom to select it; **Ctrl** or **Shift**+click adds or removes atoms, so several can be followed at once.
- **Selection Geometry** — with exactly two atoms selected, a plot of their distance over the whole trajectory appears under the MSD; with three, the angle at the middle atom (the second one clicked). Both use the minimum-image convention in each frame's own cell, share the seekable cursor with the other charts, and answer "does this bond break?" without leaving the page.
- **Track Atom** keeps the last-clicked atom centred as the run plays.
- **Periodic boundaries** default to **Unwrapped** — continuous motion, the same coordinates the MSD is measured from. Wrapping instead teleports atoms across the cell as they cross a face.

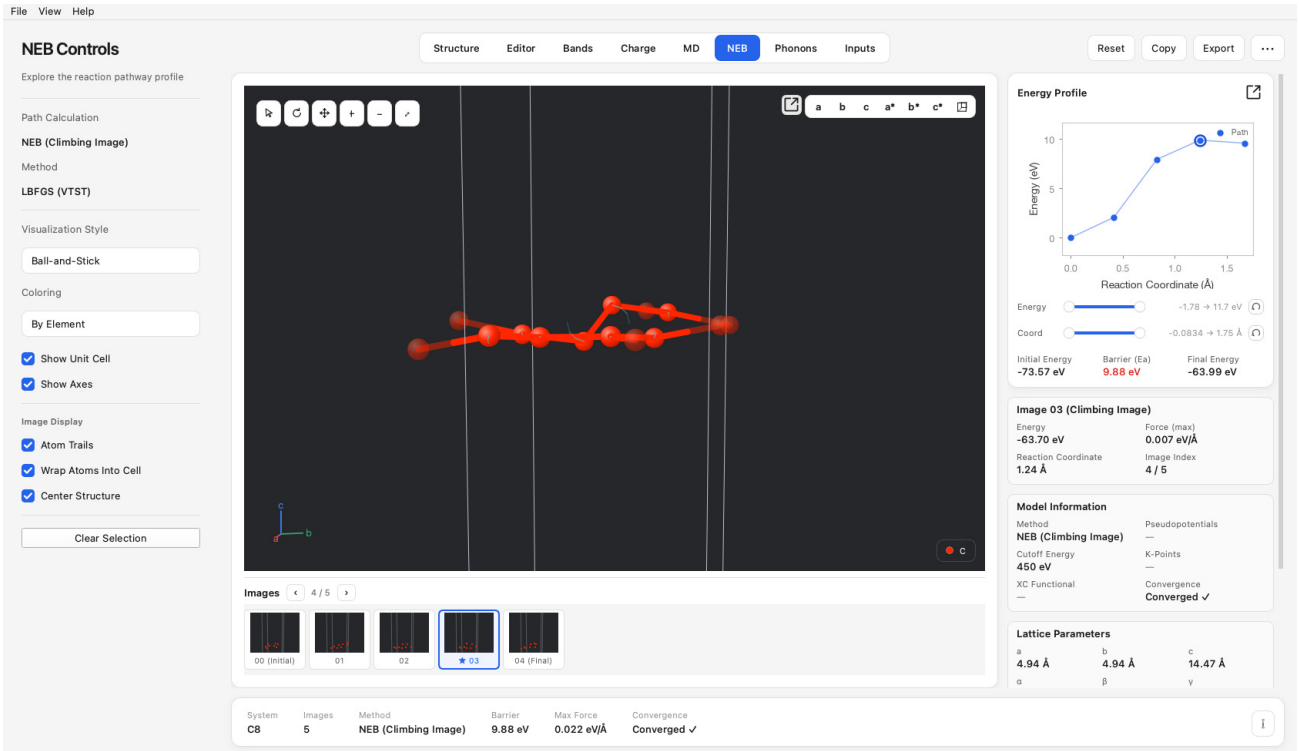
Telemetry

Temperature against its target, total/potential/kinetic energy, and mean square displacement per element. *Run Statistics* summarises the mean temperature with its spread and the energy drift — the two numbers that say whether the thermostat held and whether the timestep was sane.

A wide instantaneous temperature swing in a small cell is normal — fluctuations scale as $1/\sqrt{N}$. Read the mean, not the extremes.

NEB

Nudged elastic band: reaction path, barrier and images.



The NEB page, showing a graphene calculation.

Reads numbered image folders (00, 01, 02 …) plus their OUTCARs, or VTST's neb.dat/nebef.dat. The thumbnails, the energy profile and the 3D viewer are one selection seen three ways — clicking any of them moves the other two.

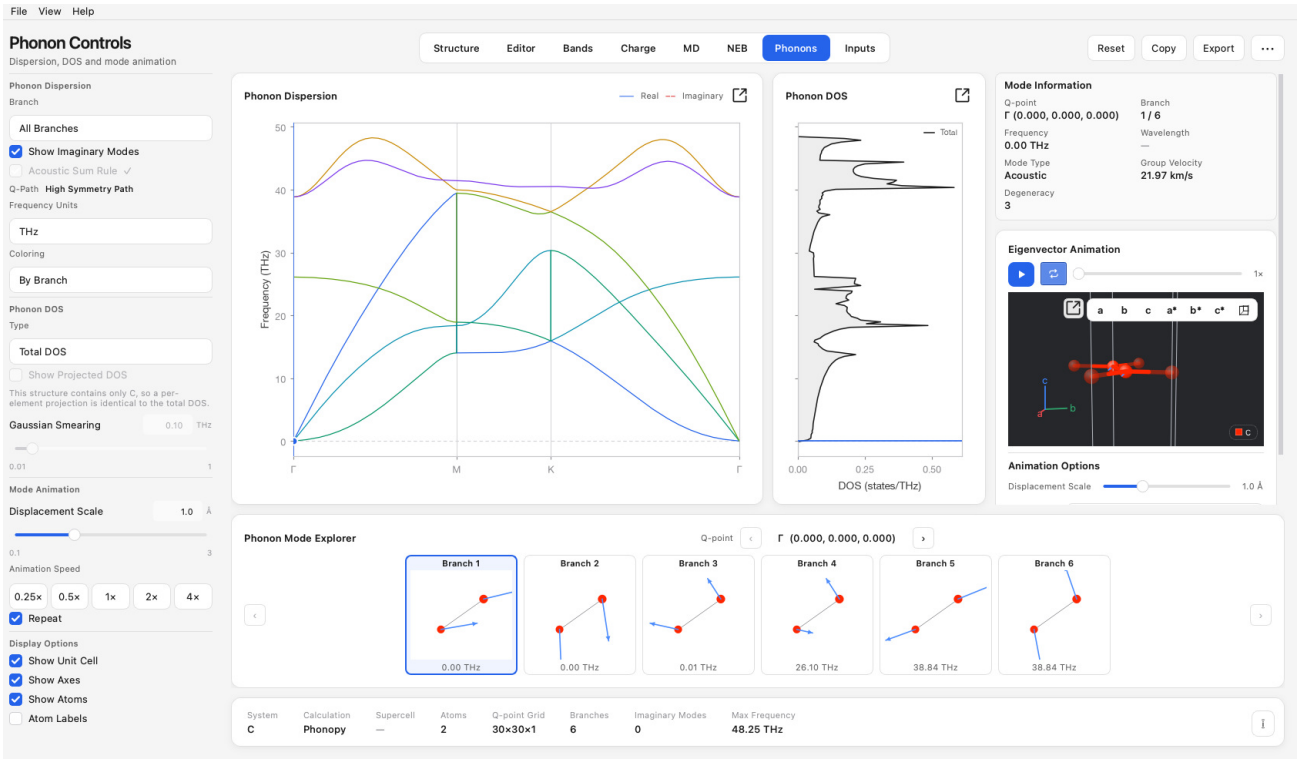
What it reports

- **Forward and reverse barriers** and the reaction energy, read from the image energies rather than assumed.
- **The saddle image** is identified and marked on the profile.
- **Atom trails** show each atom's path across the whole band, so the atoms that actually move are obvious at a glance.
- **Calculation setup** — the optimiser (VTST IOPT or VASP IBRION), spring constant and whether climbing-image was enabled.

Images whose OUTCAR is missing or unconverged are shown but flagged, rather than silently contributing a wrong energy to the barrier.

Phonons

Phonon dispersion, DOS and animated vibrational modes.



The Phonons page, showing a graphene calculation.

Reads Phonopy output (band.yaml, total_dos.dat, projected_dos.dat, phonopy.yaml), found automatically in the opened folder or a subdirectory of it.

Dispersion

Branches along the q-path, with the zero line marking the stable/unstable boundary. Click any point to select that mode.

- **By Stability** (default) draws every real branch in one colour, so instability stands out immediately.
- **By Branch** gives each branch its own colour, which lets you follow a single branch through crossings and near-degeneracies.

Red always means imaginary — a dynamically unstable mode — and is always drawn dashed, in both colouring modes. No stable branch is ever red.

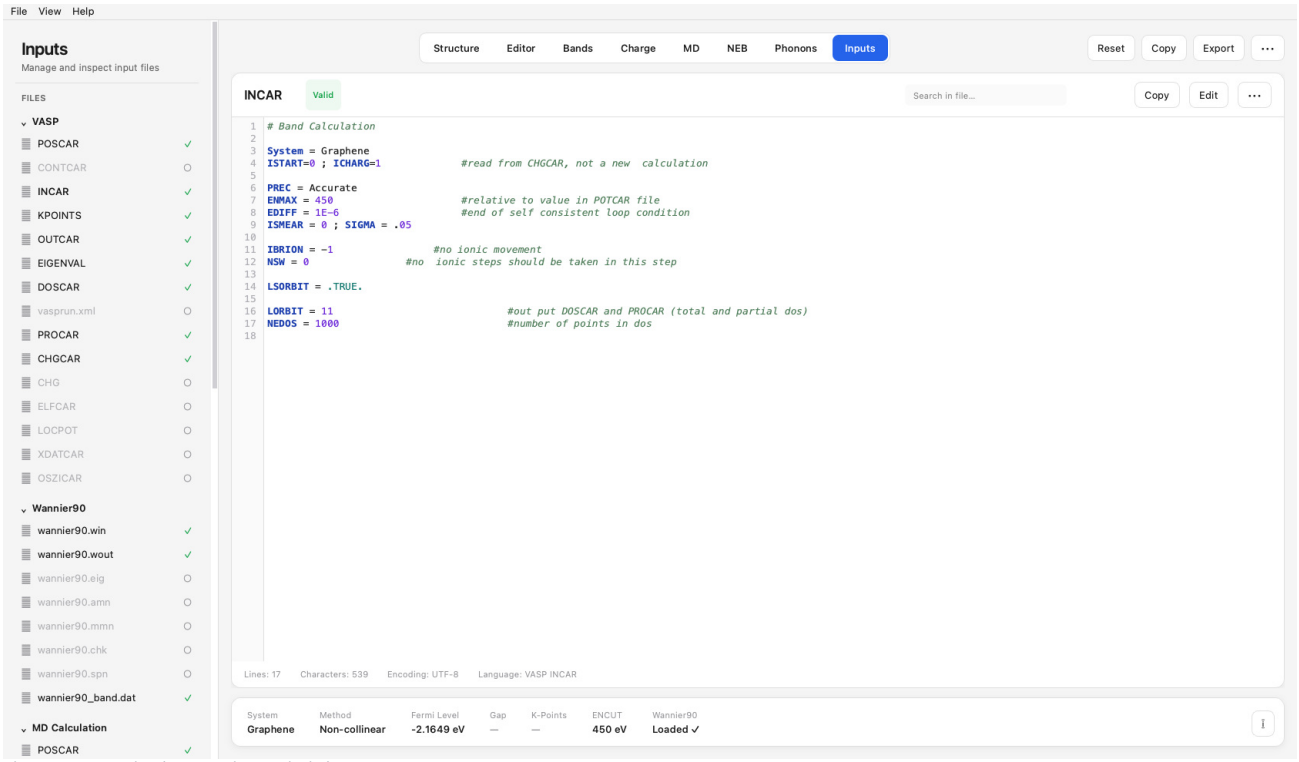
Modes and animation

Selecting a mode animates its eigenvectors on the 3D viewer: play/pause with one button, loop with the other, with adjustable displacement scale and arrow style. The acoustic sum rule is checked and reported — the three acoustic branches should reach zero at Γ.

Units switch freely between THz, cm^{-1} and meV; the data is held in THz and converted for display, so switching never rescales anything.

Inputs

Read and edit the calculation's input and output files.



The Inputs page, showing a graphene calculation.

Every recognised file in the calculation, grouped by type, with syntax highlighting for INCAR, KPOINTS and POSCAR. INCAR tags are annotated with what they mean.

- **Search in file** highlights every occurrence and reports "3 of 17"; the ‹ and › buttons step between matches, as do Enter and Shift+Enter.
- Files above a size threshold open as a preview of their first lines rather than loading a multi-gigabyte file into a text view.
- Binary files are detected and reported instead of being rendered as garbage.
- Editable files can be modified and saved in place.
- Missing files are listed too, greyed out — so you can see what a calculation *doesn't* have.

When something is missing

The application binds to *capabilities*, never to filenames. Each page asks what the calculation can support and configures itself accordingly, so an incomplete calculation degrades instead of breaking:

- A page with no data shows an empty state naming the file it needs.
- A control that cannot work is disabled with a tooltip explaining why.
- A file that exists but cannot be parsed is reported as malformed rather than silently ignored.
- An OUTCAR describing a different run than the EIGENVAL is detected and set aside, so it cannot contribute wrong metadata.

Details of any issue are available from the bottom bar, and the full log via **Help → Show Log File**.